Light

GCSE Computer Science - Character Sets (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Why do computers need character sets?

- ○ **A.** To display images on screen
- ○ **B.** To connect to the internet
- ○ **C.** To speed up the processor
- ○ **D.** To map characters to unique binary codes so text can be stored and processed

Q2 (1 mark)

When was ASCII originally developed?

- ○ **A.** The 1960s
- ○ **B.** The 1940s
- ○ **C.** The 1990s
- ○ **D.** The 2010s

Q3 (1 mark)

What is the range of character codes in 7-bit ASCII?

- ○ **A.** 0 to 127
- ○ **B.** 1 to 128
- ○ **C.** 0 to 255
- ○ **D.** 1 to 256

Q4 (1 mark)

Which types of characters are included in the ASCII character set?

- ○ **A.** Only uppercase letters
- ○ **B.** Only lowercase letters and digits
- ○ **C.** Chinese and Japanese characters
- ○ **D.** Letters, digits, punctuation marks, and control characters

Q5 (1 mark)

Approximately how many characters can Unicode represent?

- ○ **A.** 128 characters
- ○ **B.** 256 characters
- ○ **C.** Exactly 1,000 characters
- ○ **D.** Over 140,000 characters

Q6 (1 mark)

In ASCII, uppercase letters A to Z are assigned codes 65 to 90. Why is this arrangement useful?

- ○ **A.** It allows alphabetical sorting by simply comparing code values
- ○ **B.** It makes the letters appear in different colours
- ○ **C.** It reduces the file size of documents
- ○ **D.** It makes encryption impossible

Q7 (1 mark)

A computer stores the character 'Z' using ASCII. What is actually saved in memory?

- ○ **A.** The binary number representing its ASCII code
- ○ **B.** The shape of the letter Z
- ○ **C.** A sound recording of the letter

- ○ **D.** The word 'Z' written in English

Q8 (1 mark)

Which character encoding standard includes emoji (such as 😊)?

- ○ **A.** 7-bit ASCII
- ○ **B.** Extended ASCII
- ○ **C.** Binary code
- ○ **D.** Unicode

Q9 (1 mark)

In ASCII, digit characters '0' to '9' have codes 48 to 57. What code represents the digit '3'?

- ○ **A.** 3
- ○ **B.** 51
- ○ **C.** 50
- ○ **D.** 63

Q10 (1 mark)

What is a limitation of ASCII that Unicode addresses?

- ○ **A.** ASCII cannot represent any letters

- B. ASCII files cannot be saved on hard drives
- C. ASCII cannot be used on modern computers
- D. ASCII cannot represent characters from non-Latin alphabets

Q11 (1 mark)

A student is given an ASCII table. If they know the code for 'R' is 82, how can they find the code for 'S'?

- A. Divide 82 by 2
- B. Multiply 82 by 2
- C. Subtract 1 from 82
- D. Add 1 to 82

Q12 (1 mark)

Unicode was designed to be backwards compatible with ASCII. What does this mean?

- A. Unicode cannot read ASCII files
- B. ASCII was deleted when Unicode was created
- C. The first 128 Unicode codes are identical to ASCII codes
- D. Unicode uses fewer bits than ASCII

Q13 (2 marks)

In ASCII, 'a' has the code 97. Calculate the code for the letter 'k'.

- **A.** 105
- **B.** 117
- **C.** 107
- **D.** 75

Q14 (2 marks)

An ASCII code is 88. Given that 'A' = 65, which character does code 88 represent?

- **A.** X
- **B.** V
- **C.** W
- **D.** Y

Q15 (2 marks)

A text file contains the word 'HELLO' stored using 7-bit ASCII. How many bits are required?

- **A.** 5 bits
- **B.** 7 bits
- **C.** 35 bits
- **D.** 40 bits

Q16 (2 marks)

A website displays content in English, Greek, and Russian. Why must it use Unicode rather than ASCII?

- ○ **A.** ASCII websites load faster
- ○ **B.** Unicode makes websites more secure
- ○ **C.** ASCII cannot represent Greek or Russian characters
- ○ **D.** Unicode websites require less storage space

Q17 (2 marks)

In ASCII, 'N' = 78 and 'n' = 110. What mathematical relationship exists between uppercase and lowercase letter codes?

- ○ **A.** Lowercase = uppercase + 32
- ○ **B.** Lowercase = uppercase × 2
- ○ **C.** Lowercase = uppercase - 32
- ○ **D.** Lowercase = uppercase + 26

Q18 (2 marks)

The ASCII code for 'M' is 77. What is 77 in binary?

- ○ **A.** 0100 1011
- ○ **B.** 0101 1101
- ○ **C.** 0111 0111

- D. 0100 1101

Q19 (2 marks)

A document contains 200 characters. How much larger would it be if stored using 16-bit Unicode instead of 7-bit ASCII?

- A. 200 bits larger
- B. 1400 bits larger
- C. 1800 bits larger
- D. 3200 bits larger

Q20 (2 marks)

ASCII code 10 represents the 'line feed' control character. What is its purpose?

- A. To display the number 10 on screen
- B. To move the cursor to a new line in text
- C. To delete the previous character
- D. To end a computer program

Q21 (2 marks)

In ASCII, 'T' = 84. What is the code for 'W'?

- A. 85

- B. 86
- C. 87
- D. 88

Q22 (2 marks)

A scientist needs to write a document containing the Greek letter omega (Ω) and the mathematical symbol for infinity (∞). Why is Unicode essential?

- A. ASCII includes these symbols but Unicode displays them better
- B. Unicode is faster at processing mathematical documents
- C. ASCII's 128 characters do not include Greek letters or mathematical symbols like omega and infinity
- D. Unicode documents can be printed while ASCII documents cannot

Q23 (2 marks)

The ASCII code for the space character is 32. A programmer counts the number of spaces in a document by counting occurrences of which value?

- A. 0
- B. 32
- C. 48
- D. 65

Q24 (3 marks)

Calculate the sum of ASCII codes for the word 'BIG'. Given that 'A' = 65.

- ○ **A.** 207
- ○ **B.** 210
- ○ **C.** 215
- ○ **D.** 224

Q25 (3 marks)

A software developer must choose between ASCII and Unicode for a new application. Analyse the trade-offs considering: (1) storage requirements, (2) character support, and (3) compatibility with older systems.

- ○ **A.** ASCII uses less storage but only supports 128 characters and is fully compatible with older systems; Unicode uses more storage but supports global languages and is backwards compatible with ASCII for codes 0-127
- ○ **B.** Unicode is always better in every aspect with no disadvantages
- ○ **C.** ASCII supports more characters than Unicode but requires more storage
- ○ **D.** There are no practical differences between ASCII and Unicode

Q26 (3 marks)

A programmer wants to convert uppercase letters to lowercase in ASCII. Given 'A' = 65 and 'a' = 97, which operation should they apply to each uppercase character code?

- **A.** Subtract 32
- **B.** Add 32
- **C.** Multiply by 2
- **D.** Divide by 2

Q27 (3 marks)

A book chapter contains 50,000 characters. Compare the storage requirements using 7-bit ASCII versus 16-bit Unicode, expressing the difference in kilobytes (1 KB = 1000 bytes).

- **A.** ASCII: 43.75 KB, Unicode: 100 KB, difference: 56.25 KB
- **B.** ASCII: 50 KB, Unicode: 100 KB, difference: 50 KB
- **C.** ASCII: 350 KB, Unicode: 800 KB, difference: 450 KB
- **D.** Both require exactly 50 KB

Q28 (3 marks)

A programmer needs to check if an ASCII code represents a digit character ('0' to '9'). Given that '0' = 48 and '9' = 57, which condition correctly identifies digit characters?

- **A.** code >= 0 AND code <= 9
- **B.** code >= 48 AND code <= 57
- **C.** code >= 65 AND code <= 74
- **D.** code >= 97 AND code <= 106

Q29 (3 marks)

An international company is developing a customer database that must store names from customers in Japan, Saudi Arabia, Germany, and Brazil. Explain why Unicode is essential and what problems would occur using ASCII.

- ○ **A.** ASCII would work fine; Unicode is only needed for emoji
- ○ **B.** Unicode is required because ASCII can only represent Latin characters; Japanese names use kanji/hiragana/katakana, Arabic names use Arabic script, and while German and Portuguese (Brazil) use Latin characters with diacritics (ä, ü, ã, ç) that extend beyond basic ASCII
- ○ **C.** ASCII is faster, so it should be used despite the character limitations
- ○ **D.** The company should use different databases for each country to avoid encoding issues

Q30 (3 marks)

In ASCII, uppercase 'A' = 65 and lowercase 'a' = 97. A student notices that both 65 and 97 have the same remainder when divided by 32 (both give remainder 1). Analyse why this pattern exists for all letters.

- ○ **A.** This is a coincidence with no significance
- ○ **B.** The pattern only works for the letter 'A'
- ○ **C.** Both uppercase (65-90) and lowercase (97-122) letters are positioned to differ by exactly 32, so dividing by 32 gives the same remainder; this was designed to simplify case conversion in computer programs using bitwise operations
- ○ **D.** The remainder is always 0 for all letters

Total: **55** marks • Time: **30 min**

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